

KAZI FAHIM TAHMID

LECTURER, DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

UNITED INTERNATIONAL UNIVERSITY

UNITED CITY, DHAKA 1212, BANGLADESH

✉ kazifahimtahmid@gmail.com **in** linkedin.com/in/IceColdMartini  github.com/IceColdMartini

Research Interest

Enhancing **MODEL COMPRESSION METHODS AND REASONING SKILLS** to facilitate knowledge distillation while creating specialized SLMs, particularly aimed at biomedical devices, that can be deployed on end-devices, emphasizing on low-resource constraints for edge computing, ensuring privacy in medical applications, and integrating multimodal learning to bring forth precise and dependable real-time surgical and diagnostic automation.

CORE TOPICS

- MODEL COMPRESSION TECHNIQUES for SLM Development
- MULTIMODAL REASONING IN SLMs for Biomedical Data
- MACHINE LEARNING for Clinical Health and Computational Biology
- DEEP LEARNING for Biomedical Imaging and Disease Diagnostics
- DEVELOPMENT AND FINETUNING of Foundation Models within SLM scope for Multimodal Biomedical Applications

Education

Khulna University of Engineering & Technology

B.S. IN COMPUTER SCIENCE AND ENGINEERING

Khulna, Bangladesh

JAN 2020 – SEPT 2025

CGPA: 3.86/4.00 (GRADUATED AS TOP 6% IN CLASS)

- **Advisor:** PROFESSOR K. M. AZHARUL HASAN
- **Senior-year Thesis Supervisor:** PROFESSOR SK. IMRAN HOSSAIN

Awards & Grants

Dean's List Award

2021,2022,2023,2024

KUET FACULTY OF ELECTRICAL AND ELECTRONICS ENGINEERING

- Receptent of Fresher, Sophomore, Junior years for respectively Sessions 2019-2022
- Nominated for Senior year for Session 2022-23

Academic Excellence Technical Grant

2022,2023,2024

KUET DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

- Receptent for Sophomore, Junior and Senior year for Sessions 2020-23

Research Experience

Intelligent Computing Lab

RESEARCH ASSOCIATE

CSE, MIST

OCTOBER 2025 – PRESENT

RESPONSIBILITIES AND EXPERIENCE

- Developed ACTIVATION-AWARE GRADIENT-BASED QUANTIZATION WITH DISTILLATION where used joint optimization of activation quantization and distillation via gradient flow analysis to reduce activation error by a significant margin

K-MINDS

KUET MACHINE INTELLIGENCE AND DEEP LEARNING SOCIETY

RESEARCH ASSOCIATE

CSE, KUET

MAY 2025 – PRESENT

RESPONSIBILITIES AND EXPERIENCE

- Designed REPLAN-SLM, An RL-Enhanced Small Language Model for Efficient Reasoning, Tool Orchestration, and DAG-Based Task Execution
- Formulated VISION-RAG in the hyperbolic domain specifically using Poincaré disk models, to capture hierarchy better in the different radius distance from the center of the sphere [collaboration with Intelsense.AI]

Computational Intelligence & Research Group

UNDERGRADUATE RESEARCH ASSISTANT

CSE, KUET

FEB 2024 – SEPT 2025

SENIOR-YEAR RESEARCH THESIS

SUPERVISED BY: PROFESSOR SK. IMRAN HOSSAIN

RESPONSIBILITIES AND EXPERIENCE

- Formulated B-GFAN: BAYESIAN-GRAPH FOURIER ANALYSIS NETWORK, a novel uncertainty-aware Lightweight Model with proper hierarchy modelling and learnable spectral frequencies through fourier co-efficients and angular frequencies for Epilepsy Detection achieving 7x model compression (4.2M) along with better accuracy (97.8%) and F1 score(94.7%) accross patients with well-calibrated uncertainty quantification (ECE=3.1%)
- Designed ATTENTION-AUGMENTED CONVNEXT-BILSTM DUAL-BRANCH ARCHITECTURE for Hybrid EEG-Based Epileptic Seizure Detection achieving real-time (4.64 ms inference time) and achieving accuracy up to 95.76% with an F1-score of 91.25% in LOSO validation with 28M parameter count

Manuscripts in Preparation

1. SENIOR-YEAR RESEARCH THESIS: Bayesian-Graph Fourier Analysis Network: Lightweight Model for Epilepsy Detection and Comparison with Attention-Enhanced ConvNeXt-BiLSTM
EXTENDING FOR IEEE TRANSACTION SUBMISSION

2. NEURIPS 2026 SUBMISSION: Position: From Visual Plausibility to Use-Bound Validity - Governing Synthetic Data in Medical AI
WORKING ON REBUTTAL FOR ICML 2027 SUBMISSION

Teaching Experience

United International University

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
LECTURER

Dhaka, Bangladesh

JULY 2026 – PRESENT

CORE COURSES CONDUCTED

- ARTIFICIAL INTELLIGENCE (CSE 3811)
- ARTIFICIAL INTELLIGENCE LABORATORY (CSE 3812)
- MACHINE LEARNING (CSE 4889)
- DATA MINING (CSE 4891)

Military Institute of Science & Technology

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
LECTURER

Dhaka, Bangladesh

NOVEMBER 2025 – JUNE 2026

CORE COURSES CONDUCTED

- STRUCTURED PROGRAMMING LANGUAGE SESSIONAL (CSE 106): Conducted lab sessions on fundamental topics such as Conditional Logic, Iterations, Function, User-Defined Data Structures for freshers to build a introductory project of a real-life problem solver and graded the final Project Submission
- DATA STRUCTURES AND ALGORITHMS - II (CSE 215 216): Implemented advanced Graph algorithms, Dynamic Programming, and Greedy techniques in lab sessions and designed assignments on optimization of network flows and complex data structure efficiency
- DIGITAL SYSTEM DESIGN SESSIONAL (CSE 316): Instructed students on Register-Transfer Logic, Control Logic Design for them to learn computer design and designed assignments on Barrel Shifter, SAP-1 and SAP-2 (teamwise unique instruction set)
- SOFTWARE DEVELOPMENT PROJECT-I (CSE 364): Demonstrated Software Design Life Cycle (SDLC), UI/UX Design and Full-stack Development procedures in lab sessions for students to present a functional and deployable application as final project
- PATTERN RECOGNITION SESSIONAL (CSE 444): Conducted lab session on estimation techniques e.g. Classification, Regression, Clustering, Decision tree to identify different types of pattern in data

WORKSHOP

- DEEP DIVE INTO DEEP LEARNING (D3L): Introduced and conducted a 2 week-long intense theoretical and corresponding hands-on coding workshop on **RNN, Transformers, Attention Mechanism, CNN, Deep Generative Modeling, RL and LLMs** for kick-starting Junior-Year students' research journey

ADVISING

- ACTIVELY ADVISING 11 Undergraduate students of Junior Year about **personal, academic and financial** aspects to help them thrive
- INITIATED AND ARRANGED 5 focus group sessions for students of any discipline to enable **speaking out about the mental health issues they are facing** and have group discussions to find a solve

- **MENTORED** Undergrad students of KUET on **CoreML** and **Reinforcement Learning** training for Senior-year Thesis advancements
- **CONDUCTED** workshops on development of **compressed** and **multimodal LLMs** from scratch for end-device deployment
- **ORGANIZED AND MANAGED** lab session on **Model Compression** and working with **Privacy-preservation techniques** for mentees to get hands-on experience with relevant frontiers such as LoRA, Quantization, Knowledge Distillation

Industry Experience

Intelsense.AI

Dhaka, Bangladesh

SOFTWARE ENGINEER

JULY 2025 – PRESENT

RESPONSIBILITIES AND EXPERIENCE

- End-to-End ASR, TTS Development, Fine-tuning, Scaling
- Handling fast inference and load balancing for local hosting
- Agentic AI, RAG Pipelines, Agent Tool Dev & Deployment

PRODUCTS DEVELOPED

- **UNISENSEAI**, a one-stop-solution for an unified response system paired up with complete lead and transaction automation with large-scaled recommendation models
- **SENSEFORM**, a central privacy-protected form automation system for Bangladesh's national banking for home-banking and navigation with multilingual support
- **AIRVOICE**, a ASR-TTS coupled system developed for improving Huawei's Customer Service Op

Leadership

BitFest 2025

Khulna, Bangladesh

ORGANIZER

JAN, 2025

- Organized Datathon, Hackathon, LFR, Soccerbot, Poster Presentation competition segments for 12000+ undergraduate students from 150+ universities
- Participated in the dataset and task development team for preparation of Datathon and Hackathon segments
- Represented the senior undergraduate Judge in the jury panel for Hackathon and Poster Presentation

KUET Inter University Programming Contest 2025

Khulna, Bangladesh

CO-ORDINATOR

JAN, 2025

- Successfully organized an inter-university programming contest, managing participation from over 150 teams of different universities across the country.

- Coordinated a national high school science festival, engaging over 5,000 students in various activities, including Olympiads, project exhibitions, and gaming competitions

Academic Projects

CastleBlack OPENGL, C++	- Created a 3D medieval castle in OpenGL - Used Bezier curves, fractal trees, and dynamic lighting - Custom shader programming for real-time rendering and procedural generation
LangBridge FLEX, BISON, C++	- Designed a custom programming language using Flex and Bison - Included lexical, syntax, and semantic analysis
InpainterPro PYTHON, OPENCV	- Implemented an optimized Telea method for structure and texture reconstruction - Targeted restoration of missing or damaged image regions
CoinScribe SWIFT	- Developed a crypto wallet tracker app in Swift - Included features for note-taking and to-do list management - Included real-time currency exchange track charts with historical data forecasting of the stock market to make it a one-stop decision-making platform

Technical Skills

Compression	Quantization (INT8/FP16), Pruning (Structured/Unstructured), Knowledge Distillation, Low-Rank Factorization, Model Profiling & Benchmarking
Python	PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, NumPy, Pandas
MultimodalAI	Vision-Language Models, Cross-Modal Fusion, Multi-Modal Transformers, Attention Mechanisms
MLOps	Docker, K8s, Model Versioning, CI/CD Pipelines, Performance Monitoring, A/B Testing
Alignment	SFT (Supervised Fine-Tuning), DPO (Direct Preference Optimization), Knowledge Distillation for Alignment, Safety-Preserving Compression
HealthcareAI	Medical Data Preprocessing, HIPAA-Compliant Systems, Clinical Dataset Handling, Privacy-Preserving ML

English Proficiency

IELTS ACADEMIC	OVERALL BAND: 8.0
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